OSHAcademy™
Occupational Safety & Health Training

# Healthcare
## Hazardous Chemicals




Hazardous drugs, such as those used for cancer chemotherapy, antiviral treatments and hormone regimens, pose a serious health hazard to healthcare workers. This course takes a closer look at several different hazardous materials and medications that healthcare workers may be exposed to daily, as well as ways to protect themselves from serious health issues.

This page intentionally blank

# OSHAcademy Course 171 Study Guide

## Healthcare: Hazardous Chemicals

Copyright © 2022 Geigle Safety Group, Inc.

No portion of this text may be reprinted for other than personal use. Any commercial use of this document is strictly forbidden.

Contact OSHAcademy to arrange for use as a training document.

This study guide is designed to be reviewed off-line as a tool for preparation to successfully complete OSHAcademy Course 171.

We hope you enjoy the course and if you have any questions, feel free to email or call:

OSHAcademy

15220 NW Greenbrier Parkway, Suite 230
Beaverton, Oregon 97006
www.oshatrain.org
instructor@oshatrain.org
+1.888.668.9079

Disclaimer

This document does not constitute legal advice. Consult with your own company counsel for advice on compliance with all applicable state and federal regulations. Neither Geigle Safety Group, Inc., nor any of its employees, subcontractors, consultants, committees, or other assignees make any warranty or representation, either express or implied, with respect to the accuracy, completeness, or usefulness of the information contained herein, or assume any liability or responsibility for any use, or the results of such use, of any information or process disclosed in this publication. GEIGLE SAFETY GROUP, INC., DISCLAIMS ALL OTHER WARRANTIES EXPRESS OR IMPLIED INCLUDING, WITHOUT LIMITATION, ANY WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE. Taking actions suggested in this document does not guarantee that an employer, employee, operator or contractor will be in compliance with applicable regulations. Ultimately every company is responsible for determining the applicability of the information in this document to its own operations. Each employer's safety management system will be different. Mapping safety and environmental management policies, procedures, or operations using this document does not guarantee compliance regulatory requirements.

Revised April 22, 2022

This page intentionally blank

## Contents

Course Introduction ..... 1

Module 1: Hazardous Chemicals in Healthcare ..... 2

- Introduction ..... 2
- Hazardous Drug Safety and Health Plan ..... 3
  - Hierarchy of Controls ..... 4
  - Medical Services and First Aid ..... 5
- Engineering and Adminstrative Controls ..... 5
- Chemicals ..... 6
  - Ethylene Oxide ..... 6
  - Formaldehyde ..... 8
  - Glutaraldehyde ..... 8
- Health Effects ..... 9
  - Mercury ..... 10
  - Health Effects ..... 10
- Waste Anesthetic Gases ..... 12

References ..... 14

## Course Introduction

Hazardous drugs, such as those used for cancer chemotherapy, antiviral treatments, and hormone regimens, pose a serious health hazard to healthcare workers.

OSHA says even in low doses, the preparation, administration, manufacturing, and the disposal of these hazardous medications can expose hundreds of thousands of healthcare workers to irreversible harm. These hazardous medications can cause serious effects, including:

- cancer
- organ toxicity
- fertility problems
- genetic damage
- birth defects

This course discusses several different hazardous materials and medications that healthcare workers may be exposed to daily, as well as ways to protect themselves from serious health issues.

Copyright © 2022 Geigle Safety Group, Inc.

# Module 1: Hazardous Chemicals in Healthcare

## Introduction

Healthcare workers who prepare or administer hazardous drugs or who work in areas where these drugs are used may be exposed to these agents in the workplace. Some examples of hazardous drugs include those used for cancer therapy, some antiviral drugs, hormone agents, and bioengineered drugs.

Healthcare employees are also exposed to hazardous chemicals, such as pesticides and disinfectants.

About 8 million U.S. healthcare workers are potentially exposed to these hazardous drugs and chemicals.  At-risk workers include:

- pharmacy and nursing personnel
- physicians
- operating room personnel
- environmental services workers
- workers in research laboratories
- veterinary care workers
- shipping and receiving personnel
- janitorial staff

Exposure to hazardous drugs can result in adverse health effects in healthcare workers. In fact, published studies have shown that workplace exposures to hazardous drugs can cause both acute and chronic health effects.

Acute health effects usually occur rapidly as a result of short-term exposures, whereas chronic health effects generally occur as a result of long-term exposure and last longer.

Here are some examples of health effects from hazardous drugs:

- skin rashes

Copyright © 2022 Geigle Safety Group, Inc.

- adverse reproductive outcomes, including:
    - infertility
    - spontaneous abortions
    - congenital malformations
- possibly leukemia and other cancers

The health risk depends on how much exposure a worker has to these drugs and how toxic they are.

**Quiz Instructions**

After each section, there is a quiz question. Make sure to read the material in each section to discover the correct answer to these questions. Circle the correct answer. When you are finished, go online to take the final exam. This exam is open book, so you can use this study guide.

**1. The health risk of hazardous drugs depends on how much _____ a worker has to these drugs and how _____ they are.**

a. allergies, exposure
b. exposure, risky
c. risk, allergic
d. exposure, toxic

## Hazardous Drug Safety and Health Plan

OSHA has identified worker exposure to hazardous drugs as a problem of increasing health concern. When hazardous drugs are used in a healthcare facility, employers are required to implement a written program that meets the requirements of the Hazard Communication Standard (HCS). This standard provides for worker training, warning labels, and access to Safety Data Sheets (SDSs).

The HCS ensures employee awareness of the hazardous chemicals they are exposed to in the workplace.

- Agents with any of the following characteristics are considered hazardous: carcinogens, corrosives, toxic or highly toxic, irritants, sensitizers, or target organ effectors.

Copyright © 2022 Geigle Safety Group, Inc.

- Provide readily available Safety Data Sheets (SDSs) for all hazardous chemicals.
- Clearly label all hazardous chemicals, such as those found in some soaps, disinfectants, and pesticides, as hazardous. [29 CFR 1910.1200(f)]
- Train workers in potential chemical hazards and controls (engineering controls, work practices, and PPE) necessary to prevent hazards in the work area. [29 CFR 1910.1200(h)(3)]

**2. Which of these must be included as part of an employer's written health and safety program?**

a. Safety Data Sheet
b. Chemical Locator Sheet
c. Notebook of Hazards
d. Chemicals and the Worker

## Hierarchy of Controls

Workers can be protected from exposures to hazardous drugs through engineering controls, administrative controls, and personal protective equipment.

- **Engineering Controls:** Engineering controls include biological safety cabinets (BSC), compounding aseptic containment isolators, closed system transfer devices, and needleless systems. Engineering controls are the most effective way to control exposure to hazardous chemicals by preventing exposure from occurring.
- **Administrative Controls:** Administrative controls include implementing work practices, management policies, and training programs to reduce worker risk. A medical monitoring program serves as a form of secondary prevention by identifying indicators of exposure or early disease.
- **Personal Protective Equipment:** Employers are required to provide appropriate PPE, such as gloves, goggles, and splash aprons when employees are handling hazardous chemicals. [29 CFR 1910.132] PPE is the least effective control method because it relies on employee behaviors to prevent exposure.

Copyright © 2022 Geigle Safety Group, Inc.

## Medical Services and First Aid

If the eyes or body of any person may be exposed to harmful corrosive materials, suitable facilities for quick drenching or flushing of the eyes and body must be provided within the immediate area for emergency use. [29 CFR 1910.151(c)]

**3. Where should a suitable facility be located for quick flushing of the eyes and body after exposure to harmful corrosive materials in a healthcare facility?**

a. Outside the building
b. In the immediate work area
c. In another location to protect other workers
d. Close to the bathroom

### Engineering and Adminstrative Controls

Engineering and administrative controls such as an appropriate anesthetic gas scavenging system are the first line of defense and the preferred method of control to protect employees from exposure to anesthetic gases. A recent report from the Centers for Disease Control and Prevention found there is widespread use of these scavenging systems to control anesthetic gases during medical procedures. However, many other recommended controls are lacking in many facilities. Here are the results of the study:

- 3% did not always use anesthesia machines equipped with a waste gas scavenging system
- 35% started anesthetic gas flow before mask was applied to pediatric patient and 14% did the same for adult patients.
- 19% reported safe handling procedures were unavailable.
- 18% never received training on safe handling of anesthetic gases.
- 17% used a funnel-fill (open-air) system instead of a key-filler or other closed system to fill anesthesia vaporizers.
- 17% used high (fresh gas) flow anesthesia only for pediatric patients and 6% did the same for adult patients.

Copyright © 2022 Geigle Safety Group, Inc.

- 5% did not routinely check anesthesia equipment for leaks.

When precautionary practices are not followed, anesthesia care providers and bystanders are at risk of exposure. Ensuring proper precautionary measures are followed requires diligence on the part of both employers and healthcare workers.

Employers need to ensure the following:

- anesthesia care providers are initially and periodically trained
- safe-handling procedures reflecting recommended practices are in place
- anesthesia machines with scavenging systems are used (including regular inspection of anesthesia delivery equipment for leaks, periodic air and exposure monitoring prompt elimination of spills and leaks)
- medical surveillance, exposure monitoring, and other administrative controls are in place

Anesthesia care providers should seek out training, understand and follow facility procedures, ask questions, and report any safety concerns.

**4. In a recent CDC study, what percentage of healthcare workers didn't receive training on safe handling of anesthetic gases?**

a. 10%
b. 18%
c. 25%
d. 50%

## Chemicals

### Ethylene Oxide

Ethylene oxide (EtO) is used extensively by hospitals and other industries as a sterilizing agent.

EtO is a colorless, odorless gas which is both flammable and highly reactive. Most importantly, you cannot smell EtO until it reaches a temperature of 51.3 degrees Fahrenheit (10.7 degrees Celsius), the gas smells like ether.

Copyright © 2022 Geigle Safety Group, Inc.

It can be found in fumigants and sterilants and presents an opportunity for healthcare worker exposure during operations such as EtO sterilization of surgical equipment.

Human and animal studies consistently show that EtO can be hazardous to human health.

Short-term (acute) exposures to EtO can cause the following health issues:

- respiratory irritation and lung injury
- shortness of breath
- headache
- nausea
- vomiting
- diarrhea
- cyanosis (bluish or grayish color of the skin, nails, lips, or around the eyes)

Long-term (chronic) exposure over many years may cause the following:

- cancer
- reproductive effects
- genetic changes
- damage to the nervous system
- sensitization

Copyright © 2022 Geigle Safety Group, Inc.

**5. Acute exposure to Ethylene Oxide can cause which of the following health issues?**

a. Damage to the nervous system
b. Genetic changes
c. Shortness of breath
d. Cancer

## Formaldehyde

Formaldehyde is used as a disinfectant or sterilant. It is commonly used as a preservative in medical laboratories and mortuaries. Formaldehyde is also used to prepare viral vaccines, as an embalming agent, a tissue fixative, and in the sterilization of medical equipment. Paraformaldehyde, a solid polymer of formaldehyde, can be heat vaporized for the gaseous decontamination of laminar flow biologic safety cabinets. Additionally, formaldehyde is often found mixed in water and referred to as formalin.

Healthcare workers may be harmed by exposure to formaldehyde. It can irritate the skin, throat, lungs, and eyes. Repeated exposure to formaldehyde can possibly lead to cancer. The level of exposure depends on the dose, duration, and the work being done.

**6. Allison has been working in a morgue for several years and has had repeated exposure to formaldehyde. What health issue may she experience?**

a. Cancer
b. Immediate death
c. Skin rashes
d. Fatigue

## Glutaraldehyde

Glutaraldehyde is a toxic chemical that is used as a cold sterilant to disinfect and clean heat-sensitive medical, surgical, and dental equipment. Glutaraldehyde is also used as a tissue fixative in histology and pathology labs and as a hardening agent in the development of x-rays.

Glutaraldehyde is used in a limited number of applications, rather than just as a general disinfectant. Specific applications include:

Copyright © 2022 Geigle Safety Group, Inc.

- use as a disinfecting agent for respiratory therapy equipment
- bronchoscopes
- physical therapy whirlpool tubs
- surgical instruments
- anesthesia equipment parts
- x-ray tabletops
- dialyzers
- dialysis treatment equipment

## Health Effects

- **Short term (acute) effects:** Contact with glutaraldehyde liquid and vapor can severely irritate the eyes, and at higher concentrations burns the skin. Breathing glutaraldehyde can irritate the nose, throat, and respiratory tract, causing coughing and wheezing, nausea, headaches, drowsiness, nosebleeds, and dizziness.

- **Long-term (chronic) effects:** Glutaraldehyde is a sensitizer. This means some workers will become very sensitive to glutaraldehyde and have strong reactions if they are exposed to even small amounts. Workers may get sudden asthma attacks with difficult breathing, wheezing, coughing, and tightness in the chest. Prolonged exposure can cause a skin allergy and chronic eczema, and afterwards, exposure to small amounts produces severe itching and skin rashes. Glutaraldehyde has been implicated as a possible cause of occupational asthma.

**7. _____ is the main use of Glutaraldehyde.**

a. An alternative to anesthesia
b. Disinfectant for medical gloves
c. Disinfecting and cleaning heat-sensitive medical equipment
d. Cleaning patient wounds during surgery

Copyright © 2022 Geigle Safety Group, Inc.

## Mercury

Mercury is naturally occurring and exists in several forms. High mercury exposure results in permanent nervous system and kidney damage.

Employee exposure to mercury from accidental spills can occur during repair of broken thermometers, sphygmomanometers, or during sterilization and centrifugation of thermometers in maintenance areas. Mercury can also be found in some pressure-sensing instruments (e.g., barometers and sensors in machine rooms), as well as electronic equipment, and some older, medical devices. When mercury is exposed to the air, as in the case of a spill, it slowly vaporizes.

If spills are not promptly cleaned up, mercury may accumulate on surfaces and then vaporize and be inhaled by unaware workers. Mercury can also be absorbed through the skin.

### *Case Report*

A wall-mounted blood pressure unit in a doctor's office examination room failed spilling mercury on the tiled floor. The facility closed the room and hired a contractor to clean up the mercury. After cleanup, air sampling results indicated that mercury was still present. When facility staff called the State Health Department for advice, the Department conducted further air sampling and located additional beads of mercury. In all, it took the contractor several days and three cleaning cycles to completely remove all the mercury and decontaminate the office.

**8. High mercury exposure can cause which of the following health issues?**

a. Heart attacks
b. Nervous system/kidney damage
c. Imminent death
d. Cancer

## Health Effects

**Acute (short term) exposure:** Acute exposure to high levels of mercury can cause severe respiratory irritation, digestive disturbances, and marked renal damage. Acute inhalation of mercury vapor may also result in health issues, including:

Copyright © 2022 Geigle Safety Group, Inc.

- chills
- nausea
- general malaise (feeling of discomfort, illness, or uneasiness)
- tightness in the chest
- chest pains
- dyspnea (difficulty breathing or shortness of breath)
- cough
- stomatitis (inflammation of the mouth and lips)
- gingivitis
- salivation
- diarrhea

**Chronic (long term) exposure:** Chronic exposure to mercury may result in the following:

- weakness
- fatigue
- anorexia
- weight loss
- disturbance of gastrointestinal function

Copyright © 2022 Geigle Safety Group, Inc.

**9. Which of the following is listed as an acute health effect of mercury?**

a. Weight loss
b. Chest tightness
c. Weakness
d. Fatigue

## Waste Anesthetic Gases

The anesthetic gases and vapors that leak into the surrounding room during medical procedures are considered waste anesthetic gases. Healthcare professionals who work in hospitals, operating rooms, dental offices, and veterinary clinics, are potentially exposed to waste anesthetic gases and are at risk of occupational illness.

The waste anesthetic gases and vapors of concern are:

- nitrous oxide and halogenated agents (vapors) such as:
    - halothane
    - enflurane
    - isoflurane
    - desflurane

Exposure to these gases can occur through leakage of the patient's anesthetic breathing circuit during the delivery of anesthetic and through exhalations of patients recovering from anesthesia.

Some potential effects of exposure to waste anesthetic gases include the following:

- nausea
- dizziness
- headaches
- fatigue and irritability

Copyright © 2022 Geigle Safety Group, Inc.

- sterility
- miscarriages
- birth defects
- cancer
- liver and kidney disease

Employers and employees should be aware of the potential effects and be advised to take appropriate precautions.

**10. Which of the following is a potential health effect after exposure to anesthetic gases?**

a. Sneezing
b. Chest tightness
c. Coughing
d. Fatigue/irritability

Copyright © 2022 Geigle Safety Group, Inc.

## References


1. "Hazardous Drug Exposures in Health Care." Centers for Disease Control and Prevention, 4 May 2020, www.cdc.gov/niosh/topics/hazdrug/default.html.

2. "Ethylene Oxide (EtO) Understanding OSHA's Exposure Monitoring Requirements." *www.osha.gov*, www.osha.gov/Publications/ethylene_oxide.html.

3. "Hospital ETool: Healthcare Wide Hazards - Glutaraldehyde." *www.osha.gov*, www.osha.gov/SLTC/etools/hospital/hazards/glutaraldehyde/glut.html.

4. "Hospital ETool: Healthcare Wide Hazards - Hazardous Chemicals." *www.osha.gov*, www.osha.gov/SLTC/etools/hospital/hazards/hazchem/haz.html.

5. "Hospital ETool: Healthcare Wide Hazards - Mercury." *www.osha.gov*, www.osha.gov/SLTC/etools/hospital/hazards/mercury/mercury.html.

6. "Safety and Health Topics | Hazardous Drugs | Occupational Safety and Health Administration." *www.osha.gov*, 2016, www.osha.gov/SLTC/hazardousdrugs/index.html.

7. "Safety and Health Topics | Healthcare - Other Hazards | Occupational Safety and Health Administration." *www.osha.gov*, www.osha.gov/SLTC/healthcarefacilities/otherhazards.html#hazardous_drugs.


Copyright © 2022 Geigle Safety Group, Inc.